Cloud Computing

Exercise-5

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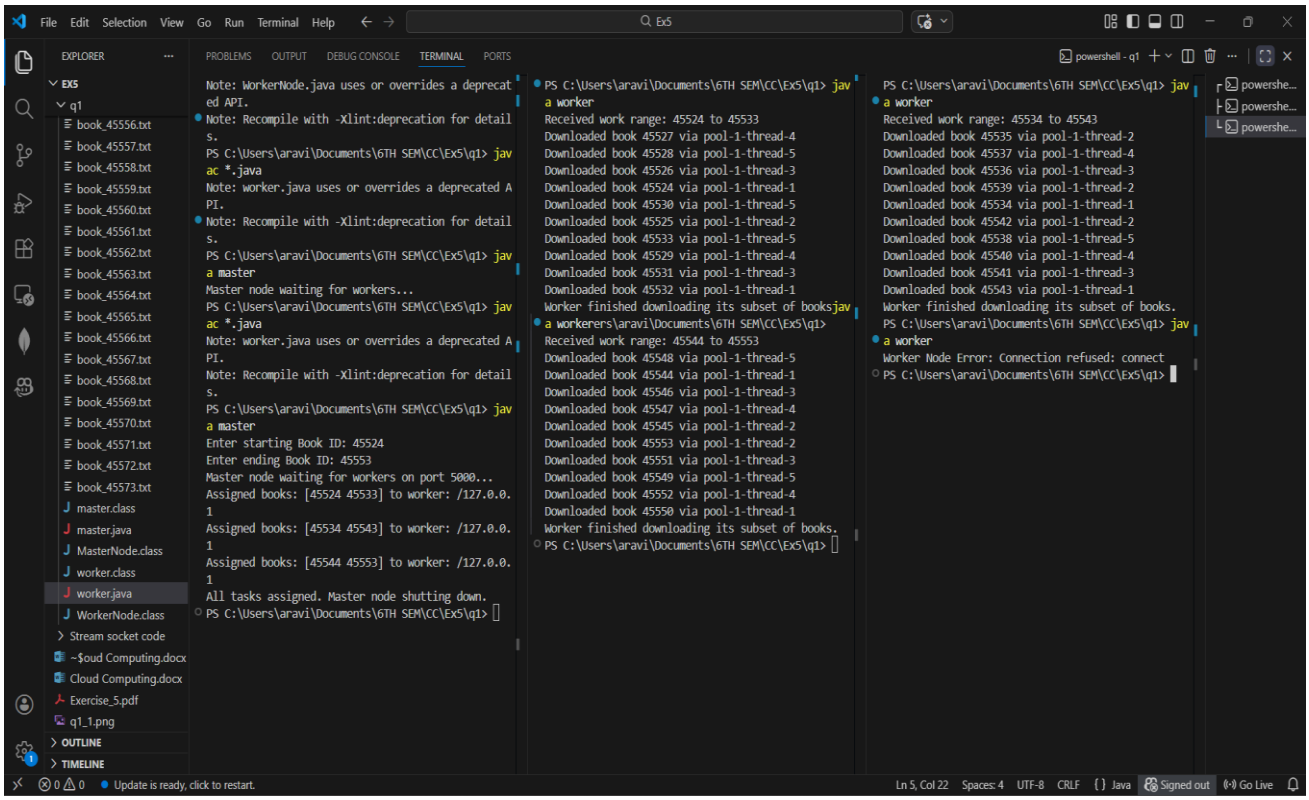
ID: 23MCCE11

Course: Imtech

1:

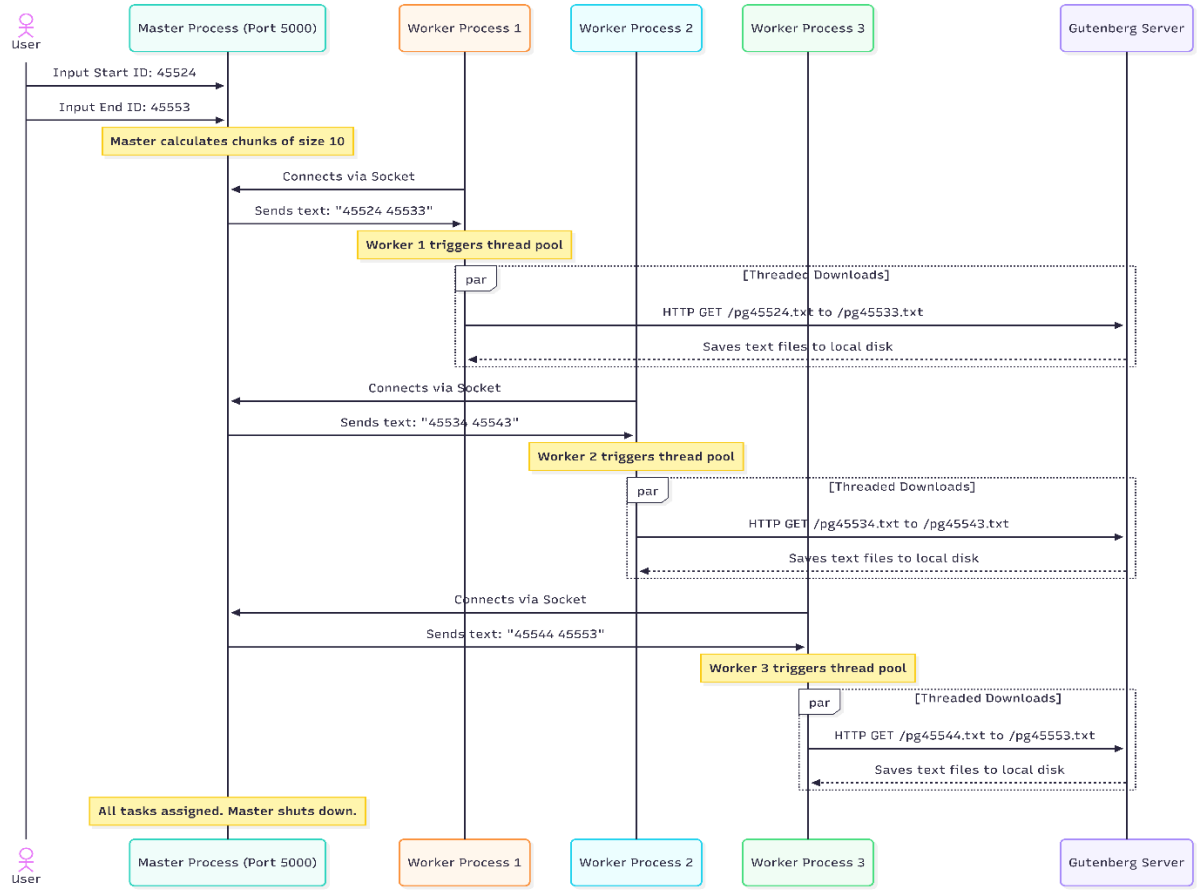
a)

Here the master will give 10 books for each worker that came and the workers will download the 10 books and ends connection. Master will wait until the all books downloaded. Worker will not wait for other workers.

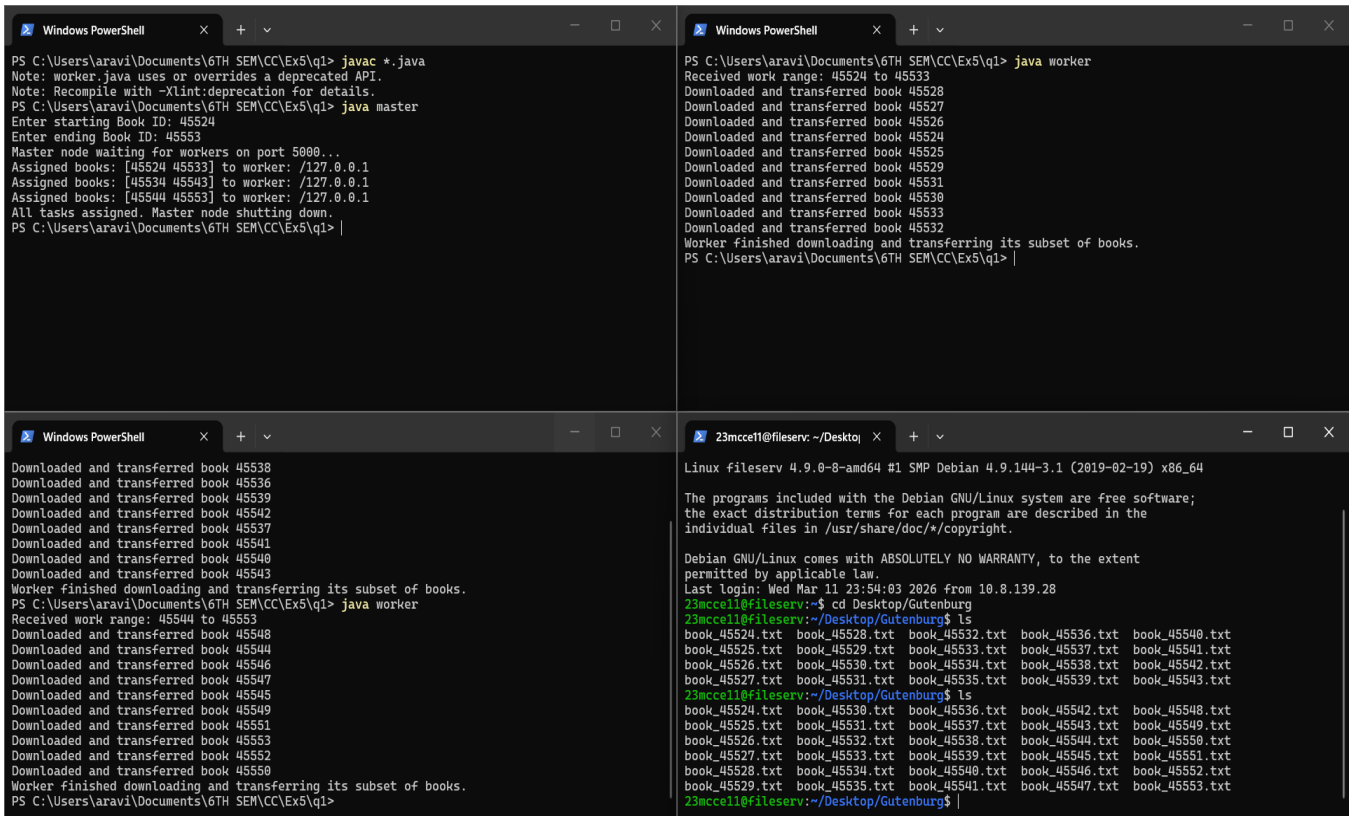


Note:

Below sequence diagram is complex and don’t think it is copied. Gpt only gave the code for the diagram in mermaid.

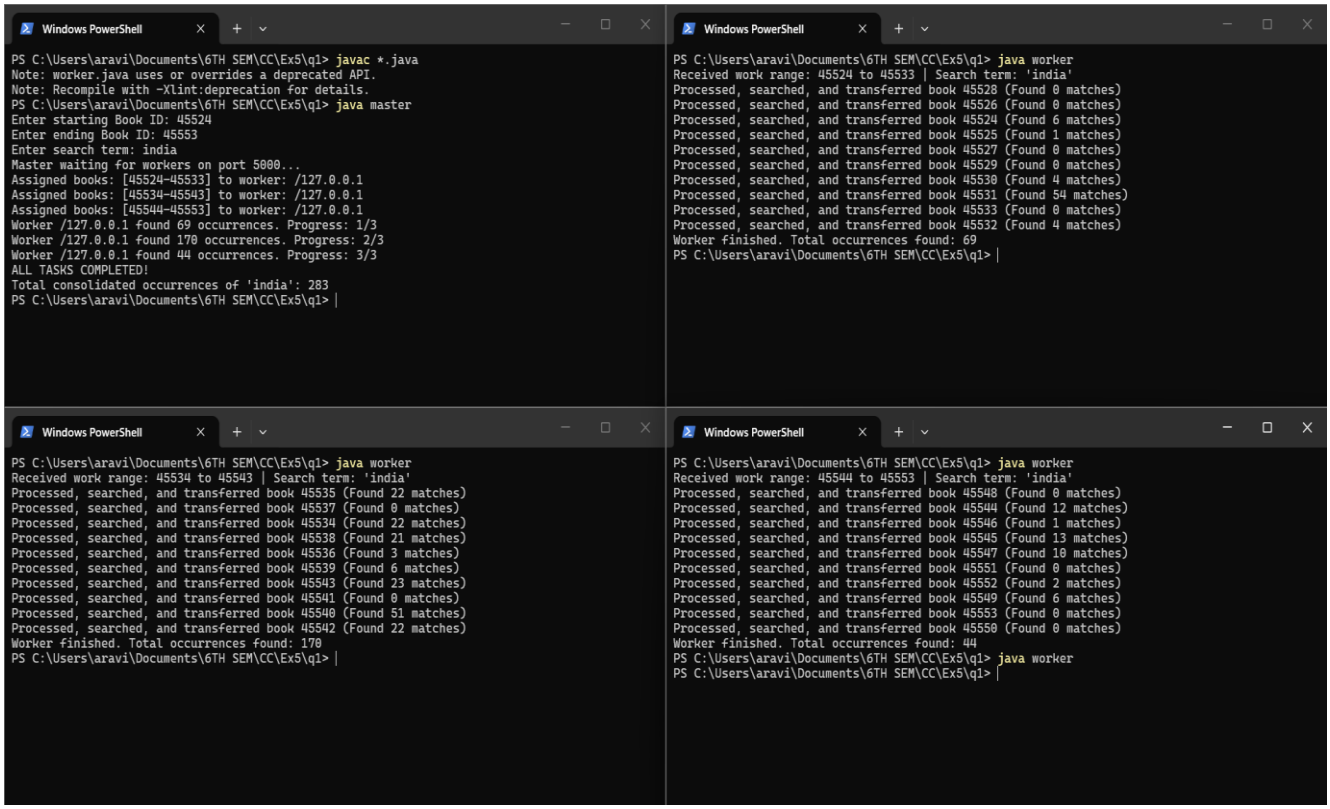

User
Master Process (Port 5000)
Worker Process 1
Worker Process 2
Worker Process 3
Gutenberg Server
Input Start ID: 45524
Input End ID: 45553
Master calculates chunks of size 10
Connects via Socket
Sends text: "45524 45533"
Worker 1 triggers thread pool
par
[Threaded Downloads]
HTTP GET /pg45524.txt to /pg45533.txt
Saves text files to local disk
Connects via Socket
Sends text: "45534 45543"
Worker 2 triggers thread pool
par
[Threaded Downloads]
HTTP GET /pg45534.txt to /pg45543.txt
Saves text files to local disk
Connects via Socket
Sends text: "45544 45553"
Worker 3 triggers thread pool
par
[Threaded Downloads]
HTTP GET /pg45544.txt to /pg45553.txt
Saves text files to local disk
All tasks assigned. Master shuts down.
User
Master Process (Port 5000)
Worker Process 1
Worker Process 2
Worker Process 3
Gutenberg Server

b)

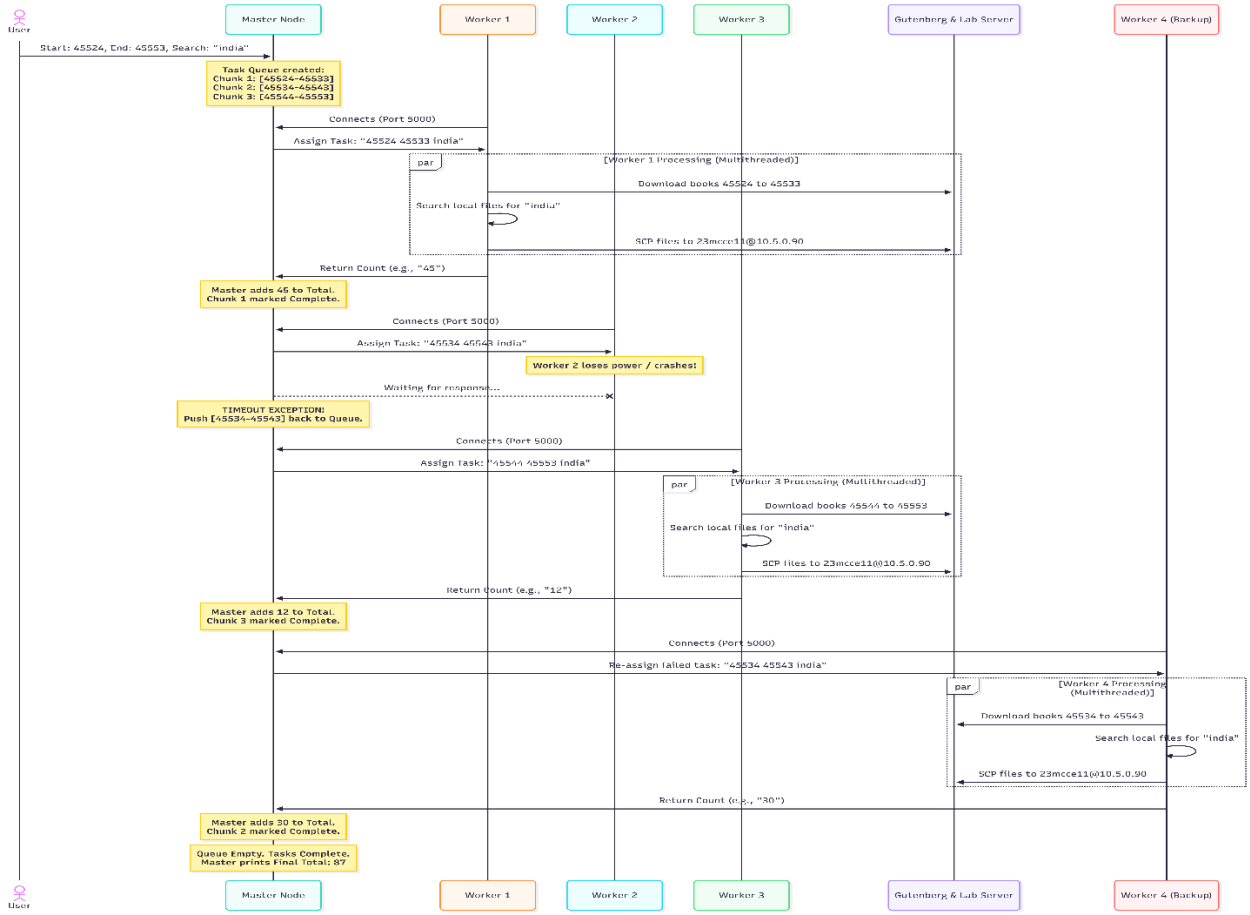

Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1> javac *.java
Note: worker.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1> java master
Enter starting Book ID: 45524
Enter ending Book ID: 45553
Master node waiting for workers on port 5000...
Assigned books: [45524 45533] to worker: /127.0.0.1
Assigned books: [45534 45543] to worker: /127.0.0.1
Assigned books: [45544 45553] to worker: /127.0.0.1
All tasks assigned. Master node shutting down.
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1>
Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1> java worker
Received work range: 45524 to 45533
Downloaded and transferred book 45528
Downloaded and transferred book 45527
Downloaded and transferred book 45526
Downloaded and transferred book 45524
Downloaded and transferred book 45525
Downloaded and transferred book 45529
Downloaded and transferred book 45531
Downloaded and transferred book 45530
Downloaded and transferred book 45533
Downloaded and transferred book 45532
Worker finished downloading and transferring its subset of books.
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1>
Windows PowerShell
Downloaded and transferred book 45538
Downloaded and transferred book 45536
Downloaded and transferred book 45539
Downloaded and transferred book 45542
Downloaded and transferred book 45537
Downloaded and transferred book 45541
Downloaded and transferred book 45540
Downloaded and transferred book 45543
Worker finished downloading and transferring its subset of books.
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1> java worker
Received work range: 45544 to 45553
Downloaded and transferred book 45548
Downloaded and transferred book 45544
Downloaded and transferred book 45546
Downloaded and transferred book 45547
Downloaded and transferred book 45545
Downloaded and transferred book 45549
Downloaded and transferred book 45551
Downloaded and transferred book 45553
Downloaded and transferred book 45552
Downloaded and transferred book 45550
Worker finished downloading and transferring its subset of books.
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q1>
23mcce11@fileserv: ~/Desktop
Linux fileserv 4.9.0-8-amd64 #1 SMP Debian 4.9.144-3.1 (2019-02-19) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 11 23:54:03 2026 from 10.8.139.28
23mcce11@fileserv:~$ cd Desktop/Gutenburg
23mcce11@fileserv:~/Desktop/Gutenburg$ ls
book_45524.txt book_45528.txt book_45532.txt book_45536.txt book_45540.txt
book_45525.txt book_45529.txt book_45533.txt book_45537.txt book_45541.txt
book_45526.txt book_45530.txt book_45534.txt book_45538.txt book_45542.txt
book_45527.txt book_45531.txt book_45535.txt book_45539.txt book_45543.txt
23mcce11@fileserv:~/Desktop/Gutenburg$ ls
book_45524.txt book_45530.txt book_45536.txt book_45542.txt book_45548.txt
book_45525.txt book_45531.txt book_45537.txt book_45543.txt book_45549.txt
book_45526.txt book_45532.txt book_45538.txt book_45544.txt book_45550.txt
book_45527.txt book_45533.txt book_45539.txt book_45545.txt book_45551.txt
book_45528.txt book_45534.txt book_45540.txt book_45546.txt book_45552.txt
book_45529.txt book_45535.txt book_45541.txt book_45547.txt book_45553.txt
23mcce11@fileserv:~/Desktop/Gutenburg$

here I created a rsa key to directly login into my server without typing password every time so that the files are transferring directly without asking password every time and we can see up to 45543 are transferred when 2 workers completed after 3rd worker completed, we can see all files.

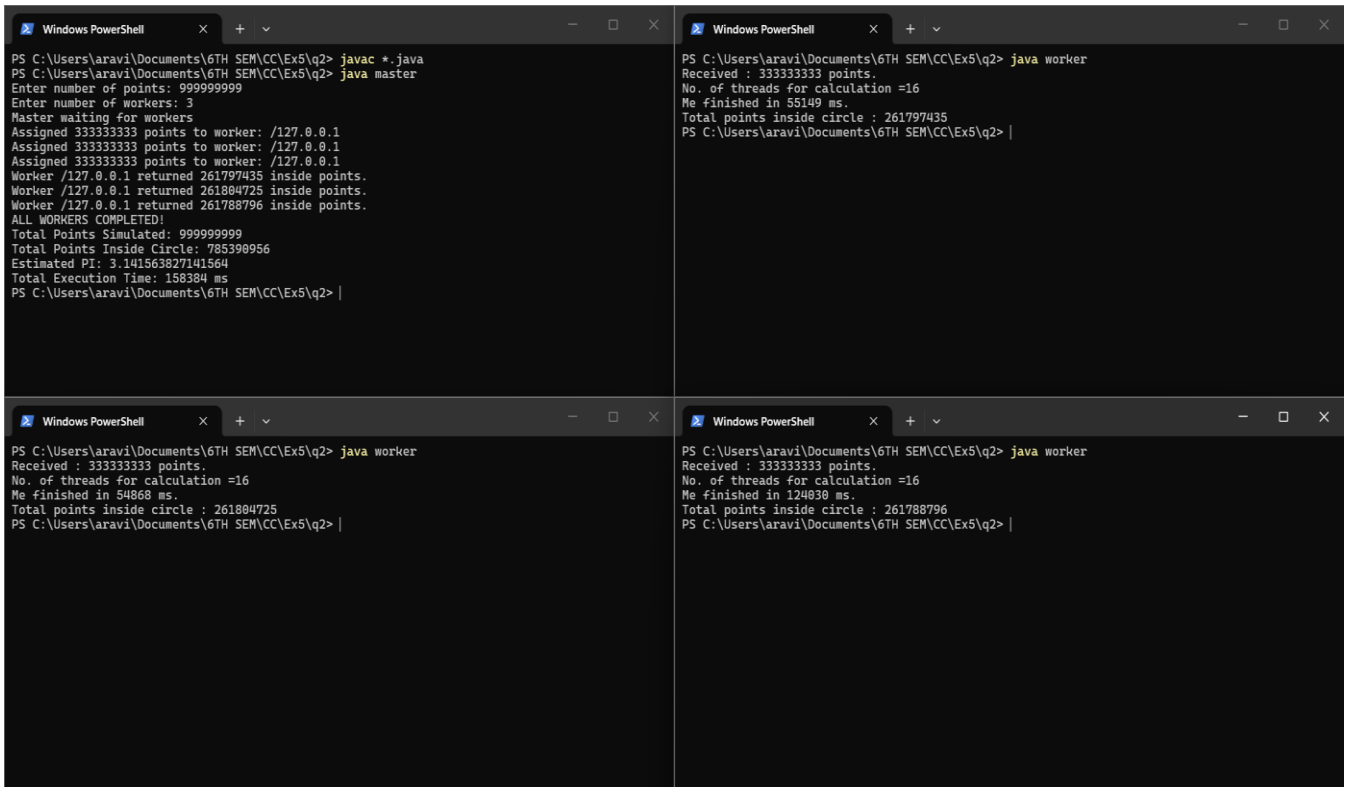
C)



Here the downloading part is same but the workers will also search the given word in the files and count it. After counting, the workers will send the number back and the master will add all the counts together. For network failures the master keeps all the id ranges in a queue, so if any worker disconnects or fails the master just takes that work and puts it back in the queue for another worker.

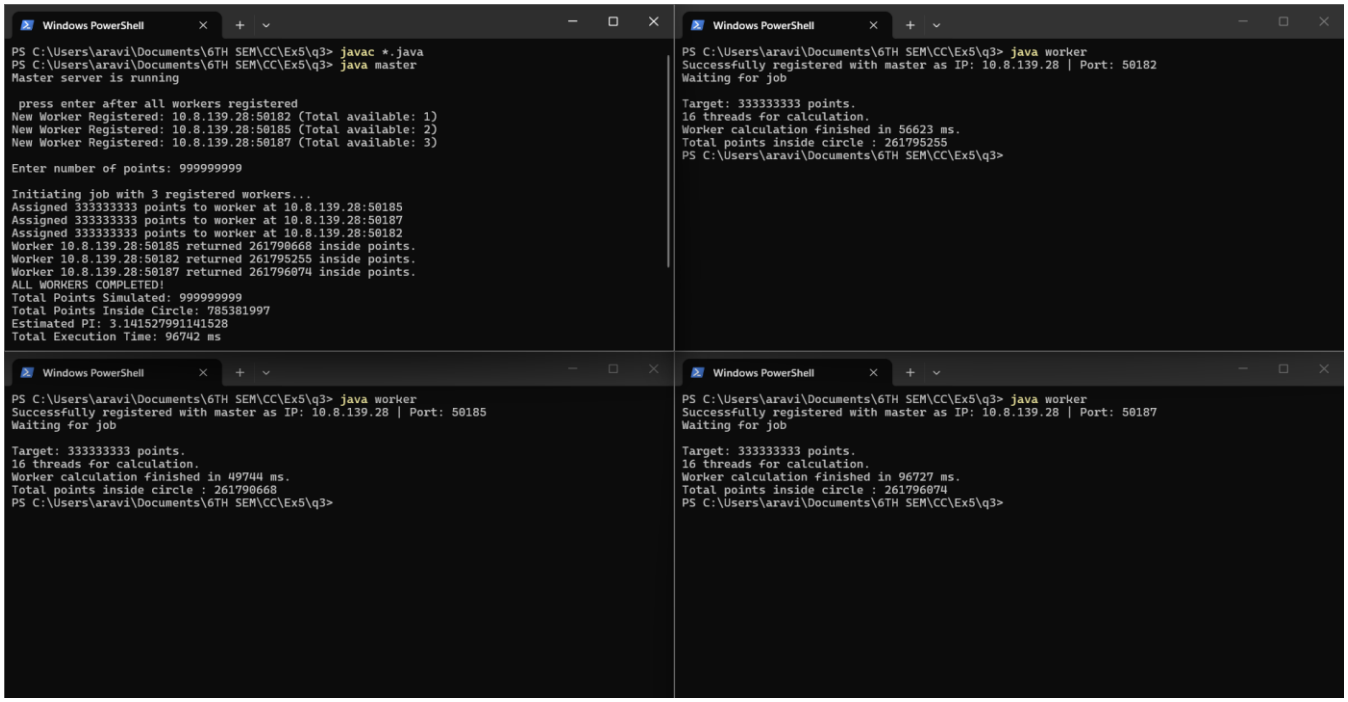

User
Master Node
Worker 1
Worker 2
Worker 3
Gutenberg & Lab Server
Worker 4 (Backup)
Start: 45524, End: 45553, Search: "india"
Task Queue created:
Chunk 1: [45524-45533]
Chunk 2: [45534-45543]
Chunk 3: [45544-45553]
Connects (Port 5000)
Assign Task: "45524 45533 India"
par
[Worker 1 Processing (Multithreaded)]
Download books 45524 to 45533
Search local files for "india"
SCP files to 23mcce11@10.5.0.90
Return Count (e.g., "45")
Master adds 45 to Total.
Chunk 1 marked Complete.
Connects (Port 5000)
Assign Task: "45534 45543 India"
Worker 2 loses power / crashes!
Waiting for response...
TIMEOUT EXCEPTION:
Push [45534-45543] back to Queue.
Connects (Port 5000)
Assign Task: "45544 45553 India"
par
[Worker 3 Processing (Multithreaded)]
Download books 45544 to 45553
Search local files for "india"
SCP files to 23mcce11@10.5.0.90
Return Count (e.g., "12")
Master adds 12 to Total.
Chunk 3 marked Complete.
Connects (Port 5000)
Re-assign failed task: "45534 45543 India"
par
[Worker 4 Processing (Multithreaded)]
Download books 45534 to 45543
Search local files for "india"
SCP files to 23mcce11@10.5.0.90
Return Count (e.g., "30")
Master adds 30 to Total.
Chunk 2 marked Complete.
Queue Empty. Tasks Complete.
Master prints Final Total: 87

2)


Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2> javac *.java
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2> java master
Enter number of points: 999999999
Enter number of workers: 3
Master waiting for workers
Assigned 333333333 points to worker: /127.0.0.1
Assigned 333333333 points to worker: /127.0.0.1
Assigned 333333333 points to worker: /127.0.0.1
Worker /127.0.0.1 returned 261797435 inside points.
Worker /127.0.0.1 returned 261804725 inside points.
Worker /127.0.0.1 returned 261788796 inside points.
ALL WORKERS COMPLETED!
Total Points Simulated: 999999999
Total Points Inside Circle: 785390956
Estimated PI: 3.141563827141564
Total Execution Time: 158384 ms
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2>
Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2> java worker
Received : 333333333 points.
No. of threads for calculation =16
Me finished in 55149 ms.
Total points inside circle : 261797435
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2>
Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2> java worker
Received : 333333333 points.
No. of threads for calculation =16
Me finished in 54868 ms.
Total points inside circle : 261804725
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2>
Windows PowerShell
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2> java worker
Received : 333333333 points.
No. of threads for calculation =16
Me finished in 124030 ms.
Total points inside circle : 261788796
PS C:\Users\aravi\Documents\6TH SEM\CC\Ex5\q2>

here we calculated 1 billion points as in EX4 it took 12.931 minutes but here it took 2.639733 minutes only.

3)



In 2 and 3 most of the working is same except for waiting and assigning ports.

In 2 the master opens a port and acts as a server waiting for workers and the workers will connect and in workers the multithreaded program will run calculating inside points

In 3 the core calculating of inside points will be same but the master waits for workers to come and exchanges their ip’s and ports, the master notes them and after master assigns work.